2017 Winter Qual

III. $A_1 + \cdots + A_k = I$

$$A_1 A_i + \cdots + A_i A_i + \cdots + A_k A_i = A_i$$

$$A_i A_1 A_i + \cdots + \cancel{A_i} + \cdots + A_i A_k A_i = \cancel{A_i}$$

$$\underbrace{x^T A_i}_{=y^T} A_1 \underbrace{A_i x}_{=y} + \cdots + x^T A_i A_k A_i x = 0, \quad x \neq 0$$

$$\Rightarrow \underbrace{y^T A_1 y}_{=0} + \cdots + \underbrace{y^T A_k y}_{=0} = 0$$

$$\Rightarrow A_i A_1 A_i = 0, \ \ldots, \ A_i A_k A_i = 0$$

$$\Rightarrow (A_1 A_i)^T (A_1 A_i) = 0, \ \ldots, \ (A_k A_i)^T (A_k A_i) = 0$$

$$\Rightarrow A_1 A_i = 0, \ \ldots, \ A_k A_i = 0$$

$$\Rightarrow A_i A_j = 0 \quad \forall i \neq j$$

∎

$\boxed{2}$.

$$S_{xy} = \frac{x^T(I-H_1)y}{n-1}$$

$$\begin{aligned}\mathbb{E}[x^T(I-H_1)y] &= tr((I-H_1)\sigma_{xy}) + \mu_x(I-H_1)\mu_y \\ &= \sigma_{xy}(n-1) + \underbrace{\mu_x 1(I-H_1)1\mu_y}_{=0} \\ &= \sigma_{xy}(n-1)\end{aligned}$$

$$\mathbb{E}[S_{xy}/n-1] = \sigma_{xy} \quad : \text{UE}$$

$\blacksquare$

$\boxed{4}$.

$$\theta = X\beta + u, \quad Y = X\beta + u + e$$

$$\hat{\theta}^{BLUP} = \widehat{x_0\beta}^{BLUE} + \mathrm{cov}(\theta, Y)\,\mathrm{cov}(Y)^{-1}(Y - \widehat{X\beta}^{BLUE})$$

$$\mathrm{cov}(\theta, Y) = \mathrm{cov}(x\beta + u, \theta + e)$$
$$= \mathrm{cov}(x\beta + u, x\beta + u + e)$$
$$= \mathrm{var}(u) + \mathrm{cov}(u, e)$$
$$= \sigma I_m$$

$$\mathrm{cov}(Y) = \mathrm{cov}(x\beta + u + e)$$
$$= \sigma I_m + S$$

$$\hat{\theta}_i = x_i\tilde{\beta} + \frac{\sigma}{\sigma + S_i}(y_i - x_i\tilde{\beta})$$

$$= \frac{\sigma}{\sigma + S_i} y_i + \frac{S_i}{\sigma + S_i} x_i\tilde{\beta}$$
$$= (1 - \lambda_i) y_i + \lambda_i x_i \tilde{\beta}$$

$\tilde{\beta}$ is BLUE? , $\mathrm{Var}(Y) = \sigma I_m + S =: V$

$$Y \sim N(X\beta, V) \Rightarrow V^{-1/2} Y = Y^*$$

$$\Rightarrow \frac{\partial \| y^* - X^*\beta \|^2}{\partial \beta} = -2X^{*T}(y^* - X^*\beta) = 0$$

$$\Rightarrow \tilde{\beta} = (X^{*T}X^*)^{-1} X^{*T} y^*$$

$$= (X^T V^{-1} X)^{-1} X^T V^{-1} y \ ; \ \text{BLUE}$$

(b) $\operatorname{var}(y) = \operatorname{var}(X\beta + u + e)$

$$= \sigma I + I$$

$$= (\sigma + 1) I$$

$$\mathbb{E} Y = X\beta$$

$$\mathbb{E}[y^T (I - H_X) y] = tr((I - H_X)(\sigma+1) I) + 0$$

$$= (m - p)(\sigma + 1)$$

$$\frac{y^T (I - H_X) y}{m - p} - 1 \ ; \ \text{UE of } \sigma$$

$\boxtimes$